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Bài tập ngày 28-4

Chủ đề: Lập trình ứng dụng server thực hiện chức năng định tuyến dữ liệu đơn giản như sau:

+ Server chờ các kết nối ở cổng 9000.

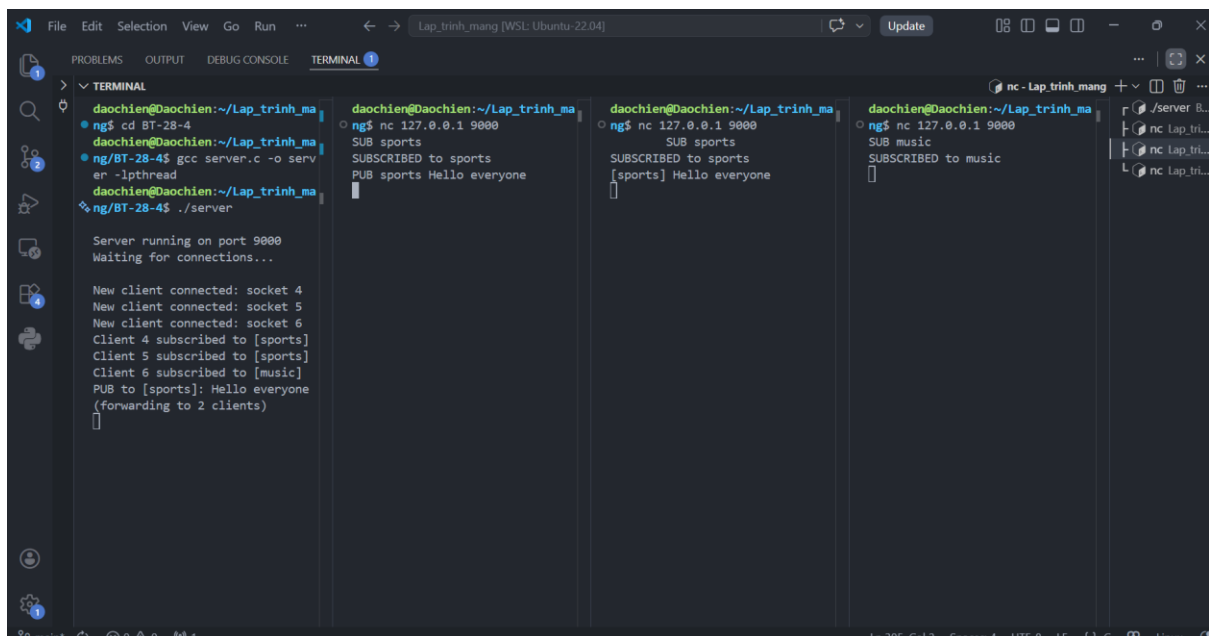
+ Client gửi “SUB <topic>” để đăng ký nhận tin nhắn thuộc chủ đề topic.

+ Client gửi “UNSUB <topic>” để hủy đăng ký nhận tin nhắn thuộc chủ đề topic.

+ Client gửi “PUB <topic> <msg>” để gửi tin nhắn **msg** đến chủ đề **topic**.

+ Server nhận tin nhắn **msg** và chuyển tiếp cho tất cả các client đã đăng ký chủ đề **topic** trước đó.

Chú ý: Một client có thể đăng ký cùng một lúc nhiều chủ đề khác nhau.



```
daochien@Daochien:~/Lap_trinh_ma$ ng$ cd BT-28-4
daochien@Daochien:~/Lap_trinh_ma$ ng/BT-28-4$ gcc server.c -o server -lpthread
daochien@Daochien:~/Lap_trinh_ma$ ng/BT-28-4$ ./server

Server running on port 9000
Waiting for connections...

New client connected: socket 4
New client connected: socket 5
New client connected: socket 6
client 4 subscribed to [sports]
client 5 subscribed to [sports]
client 6 subscribed to [music]
PUB to [sports]: Hello everyone
(forwarding to 2 clients)

daochien@Daochien:~/Lap_trinh_ma$ ng$ nc 127.0.0.1 9000
SUB sports
SUBSCRIBED to sports
PUB sports Hello everyone

daochien@Daochien:~/Lap_trinh_ma$ ng$ nc 127.0.0.1 9000
SUB sports
SUBSCRIBED to sports
[sports] Hello everyone

daochien@Daochien:~/Lap_trinh_ma$ ng$ nc 127.0.0.1 9000
SUB music
SUBSCRIBED to music
```

- Server startup, 3 client kết nối, Client A, B sub sports; C sub music

```
daochien@Daochien:~/Lap_trinh_ma$ ng$ cd BT-28-4
daochien@Daochien:~/Lap_trinh_ma$ ng/BT-28-4$ gcc server.c -o server -lpthread
daochien@Daochien:~/Lap_trinh_ma$ ng/BT-28-4$ ./server
Server running on port 9000
Waiting for connections...

New client connected: socket 4
New client connected: socket 5
New client connected: socket 6
Client 4 subscribed to [sports]
Client 5 subscribed to [sports]
Client 6 subscribed to [music]
PUB to [sports]: Hello everyone
(forwarding to 2 clients)
Client 4 subscribed to [music]
PUB to [music]: Rock and roll (forwarding to 2 clients)
[ ]
```

- A pub sports → B nhận, C pub music → A nhận (vì A đã sub music)
-

```
daochien@Daochien:~/Lap_trinh_ma$ ng$ cd BT-28-4
daochien@Daochien:~/Lap_trinh_ma$ ng/BT-28-4$ gcc server.c -o server -lpthread
daochien@Daochien:~/Lap_trinh_ma$ ng/BT-28-4$ ./server
Server running on port 9000
Waiting for connections...

New client connected: socket 4
New client connected: socket 5
New client connected: socket 6
Client 4 subscribed to [sports]
Client 5 subscribed to [sports]
Client 6 subscribed to [music]
PUB to [sports]: Hello everyone
(forwarding to 2 clients)
Client 4 subscribed to [music]
PUB to [music]: Rock and roll (forwarding to 2 clients)
Client 4 unsubscribed from [sports]
PUB to [sports]: Only B now (forwarding to 1 clients)
[ ]
```

- A unsub sports, B pub sports → chỉ B nhận

```
File Edit Selection View Go Run ... Lap_trinh_mang [WSL: Ubuntu-22.04] Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
daochien@Daochien:~/Lap_trinh_ma
ng$ cd BT-28-4
daochien@Daochien:~/Lap_trinh_ma
ng/BT-28-4$ gcc server.c -o server -lpthread
daochien@Daochien:~/Lap_trinh_ma
ng/BT-28-4$ ./server

Server running on port 9000
Waiting for connections...

New client connected: socket 4
New client connected: socket 5
New client connected: socket 6
Client 4 subscribed to [sports]
Client 5 subscribed to [sports]
Client 6 subscribed to [music]
PUB to [sports]: Hello everyone
(forwarding to 2 clients)
Client 4 subscribed to [music]
PUB to [music]: Rock and roll (forwarding to 2 clients)
Client 4 unsubscribed from [sports]
PUB to [sports]: Only B now (forwarding to 1 clients)
[]

daochien@Daochien:~/Lap_trinh_ma
ng$ nc 127.0.0.1 9000
SUB sports
SUBSCRIBED to sports
PUB sports Hello everyone
SUB music
SUBSCRIBED to music
[music] Rock and roll
UNSUB sports
UNSUBSCRIBED from sports
[]

daochien@Daochien:~/Lap_trinh_ma
ng$ nc 127.0.0.1 9000
SUB sports
SUBSCRIBED to sports
[sports] Hello everyone
PUB sports Only B now
[]

daochien@Daochien:~/Lap_trinh_ma
ng$ nc 127.0.0.1 9000
SUB music
SUBSCRIBED to music
PUB music Rock and roll
PUB movie No one hears
Topic [movie] has no subscribers
UNSUB movie
Not subscribed to movie
[]
```

- PUB topic không ai sub → báo lỗi, UNSUB topic chưa sub → báo lỗi